

A Fast Universal Collision-free Agentic Model: Compact Illusory Representation and Memory-Efficient Incremental Mapping.

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<https://github.com/robotsorcerer/denseslam>

Abstract—We present a generalist collision-free computational agent that rapidly embeds acquired knowledge about real world environments into its internal model or “head” based on a pipeline of machine learning ensembles wrapped around a sequence of GPU-accelerated approximate convex decomposition, a probabilistic convex set polytopes computational scheme, frontier-based planning schemes, and low-level non-linear control for general computational geometry navigation tasks. The agent’s head stores a compact, memory-efficient, and computationally tractable internal model of the environment that *proactively* constructs a collision-free model based on exogenous perception, updates and maintains acquired state, whilst adaptively modifying erstwhile computed states based on newly retrieved information from external stimuli. This agentic design offers flexibility in many real-time applications and encourages self-collision awareness, rapid fault diagnosis and recovery in complex environments, and provides an efficient storage mechanism that makes it suited to long-range mapping and tracking systems. We train the agent on diverse dataset and leverage machine learning in reasoning through an ensemble of established and novel computational geometry, control systems, and convex optimization algorithms. Such a model is increasingly becoming a necessity in many ‘embodied agentic’ and emergent real-time AI applications. This work serves as a first step in answering the call for a foundational generalist agent that uses all perception information available for its autonomous exploration (vision, tactile, ranging etc) in rapidly building a *correct and minimal* environment representation that can then be leveraged for further downstream tasks.

I. INTRODUCTION

A. Context and Motivation

The lack of a fast universally stable, and reliable *autonomous navigation engine* for the real-time illusory processing of information from perceptual sensory devices [12] that is generalizable, and memory and computationally efficient has been a limitation in the wide adoption of the several splintered efforts in foundation models [1, 2, 6, 8, 24, 44, 55, 57] or established planning algorithms [17, 27, 34, 47, 63] for real-time and fast *perception engineering* [51] within autonomous navigation settings in diverse everyday *embodied* ‘agentic’ applications.

Our ‘model’ taxonomy follows Roth-Tabak and Jain [71]’s description, that is, an agent’s power of representation rather

than its specific model instances that may lack completeness or correctness. The internal model principle in neuroscience [16] is a splendid inspiration in our work for modeling autonomous reasoning agents with *compact, memory-efficient internal information on the external world*. In this internal model principle, if an agent possesses a *minimum model* of “external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilize the knowledge of past events in dealing with the present and the future, and in every way to react in a much fuller, safer, and more competent manner to the emergencies which face it” [16]. With Craik [16]’s view that the nervous system is a calculating machine with a modeling or paralleling capacity for external events that engenders thought and of explanation [16], we seek to build agents with the representation power above that generalize into the diverse tasks earlier enumerated.

Large deep models and policies have emerged as viable mechanisms for encoding such perceptual experiences and building these agentic systems [88]. Two application categories have broadly emerged in these large agentic paradigms: (1) situated agents i.e. agents trained on stored information in the form of static data albeit without an embodied actuation such as proprioception or sensing mechanism such as tactile, or vision perception; and (2) embodied agents i.e. agents trained on signals from experiential and interactive feedback from proprioception, kinesthesia, vision, or other three-dimensional sensors.

Text-based question-answering autonomous agents [64, 20], vision [44], or vision-language agents [65, 2, 55] are situated because they interact with the world through received information. On the other hand, a field robot that incrementally learns about the abstract representation of the world via interaction is embodied but not situated. Models trained along this second category may include multimodal vision-language-action (VLA) models [1, 8, 24]. Often lumped together with situated models as foundation models [6], they have improved the capability of large language assistants in self-driving [54] and supposedly automated tasks in general [8, 1, 57], prompting the abduction that they might be the elixir for automating complex automation problems in gaming, robotics, physics simulations, virtual reality, animations, and in embodied devices. However, there are challenges that these frontiers present that requires a

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closer application of thinking and design to build agents that are truly autonomous.

We label a modeling engine capable of embodied interaction an “*agent*”. Such an agent may be virtual, or an embodied physical machine such as a head mounting device (HMD) or robot. These have an ability to alter the environment based on internal actuation (proprioception) in response to external stimuli, perceived via e.g. kinesthesia, vision, or tactile sensing. The applications of such agents concern several ‘*agentic*’ interactive applications including mobile robots, (mobile) robot manipulators, autonomous vehicles, or video game non-player characters (NPCs) *inter alia*.

Usually trained on situated data, they hardly provide stability and safe predictability in embodied applications, especially physical ones. This has hampered their use in industrial settings and called into question whether a generalist and universal model can be trained solely from large-scale data that allows principled compositional reasoning for a correct and complete model of the physics of interconnected mechanisms that constitute modern robots, head-mounting devices, mechanical machines and other embodied agents. Towards a universal model that can solve embodied modeling and control problems such as envisioned in efforts including OpenVLA [43], PaLM-E [24], RT-2 [8] among others.

B. The case for an embodied agentic navigation model

A complete systems specification for a universal agentic navigation model would include a capacity for incremental situational awareness in obstacle-cluttered complex environments that may require long-range mapping (without the limitations of memory management) and be agnostic to airborne, underwater, (extra-) terrestrial, and augmented (including virtual, mixed, and simulated) reality applications. And application requirements may involve dynamic two- or three-dimensional (2D/3D) sensing (before the compact representation of illusory motion is computed) for embodied planning.

In this sentiment, a universal agent that fulfills these requirements calls for a machine learning model built out of several computationally efficient pipelines: computational geometry [68], convex decomposition [56], semidefinite programming [7], and general planning schemes [49]. Ideally, such an agent would be trained on diverse large-scale datasets with these ensemble of *modular* algorithms that enable compact representation of collision-free space in many real-time applications.

A computationally bulky aspect of this machine learning ensemble would be creating a computationally compact representation of the environment for interactive agents that can encode the *stroboscopic apparent motion* [69] illusory effects of perceptual experiences under the characteristic uncertainty that plagues most digital storage means (such as range sensors, RGB, depth, or event cameras among others).

C. Mapping Minimalism: Beyond Dense SLAM

The next frontier in *autonomous embodied intelligence* is to build truly intelligent automatons capable of embodied

planning and control in noisy and uncertain two- or three-dimensional perceptual environments [19] without cumbersome representations such as is common with modern spatial semantic mapping and tracking systems [36, 37]. We would expect such an engine to possess a fast throughput, low latency, stable predictions and decisions that are resilient in the face of uncertainty.

The dense simultaneous localization and mapping (SLAM) outlined in Davison [19] and now being realized for indoor scenes in Hughes et al. [37], incrementally builds a map based on 3D semantic spatial labels and representations and location-specific geospatial information as the agent navigates an environment whilst determining its own pose within the map. It uses a hybrid of hand-engineered and learning-based mixture of algorithms to build a neural SLAM [81] engine. Other efforts leverage neural fields [59] or 3D Gaussian splatting [42] alongside a continuous and differentiable scene model in generating an improved memory efficiency and robust measurement estimation in navigation settings; however, these are computationally exacting in resource memory, requiring an avalanche of training data combined with precise ground truth annotations — thus, limiting their proficiency in previously unseen environments.

An ideal goal of an intelligent neural SLAM agent would be to operate on the time-scales of intelligent creatures such as animals and human beings by providing a continuous model of surfaces, minimize memory requirements, improve robustness to noise or outlier processing, mitigate against occlusions and provide compact and informative maps. In addition, such a scheme would reduce the computational requirements needed in the backend optimization so that the neural SLAM agent provides: agnosticism to environment size, support mission time growth, improve memory efficiency, and inculcate robust pose estimates whilst enhancing real-time performance. Building agents capable of these present a training challenge on diverse large scene datasets of embodied agents, employ minimal computational power, manage memory requirements, and imbue robustness into its structural representation so that its latent space do not collapse.

It begs to ask, if the ultimate goal of an agent is to autonomously explore an environment, would it make sense to invest too much computational resources in a full 3D reconstruction of the environment? In SLAM pipelines, the process of an agent *tracking* its pose within an environment is termed “*localization*”; and the conventional wisdom literature is to first build a map before the localization phase. 3D mapping, while a critical component of many SLAM pipelines, is a costly computational process. It is seldom available before the tracking process starts. This means in large-scale navigation settings, in unstructured environments, the memory requirements required for 3D reconstruction is often a bane for the applications of *in-situ* algorithms that traditionally tackle the SLAM problem. It is even more challenging in emergent DenseSLAM [80] applications where the goal is to incorporate abstract reasoning modules into the SLAM engine [35, 70] such as helping an agent locate what

is where within a large map, often using natural language, semantic spatial information or the likes.

Since the purpose of the map is to have a good localization efficacy, it is worth revisiting the computational pipeline in 3D spatial SLAM if full 3D reconstruction is at all a necessary requirement and if there are gains to be made by leveraging machine learning alongside computational geometry algorithms and convex optimization and decomposition mechanisms. This is the essence of this paper. Therefore, we consider an ensemble subsystems and ideas into a mosaic of an AI ‘*model*’ that emphasizes the need to learn collision-free volume in the environment first; and then the task of localization is built upon this.

The applications of the internal model principle cut across disciplines including biology [4], control theory [39], consciousness [74], and even sensorimotor control [87]. It is a redounding inspirational source behind our work. To herald the technological agentic systems that can create rapidly deconstruct cluttered scenes, delineate empty space from occupied, and unknown spaces within the map of their world, agents with memory and computational efficacy on unstructured large-scale practical problems are trained with machine learning wrapped around mathematical models of what is understood. This is crucial for analyzing and rapidly predicting consequences in most engineering and application protocols. In emphasis and focus, we create a model that represents free-space (CFree) and unknown, unmapped locations in complex diverse scenes in an incremental and relative quick succession, with our eye towards runtime and memory efficiency in many real-time applications. Succinctly, our **contributions** are delineated as follows:

- First, rather than reconstruct an entire scene, we focus on reconstructing the environment’s collision-free space in a volumetric manner. This presents huge cost savings in terms of using a map for planning. All obstacles are abstracted implicitly in a coarser map.
- Within this collision-free space computational learning scheme, we seek to reduce storage and free-space computational costs; therefore, we introduce an efficient approximate convex decomposition of the scene before the back-optimization stage;
- Seeing we want a model that can generalize to large environments and improve mission-critical real-time computational challenges, we train a model on diverse datasets via graph-based convex programming (to engender tractable obstacle-free computation) on open-source three-dimensional datasets.

We focus on three-dimensional (3D) vision-enabled planning and control systems (often within hazardous environments). In these situations, wrong (model) predictions or mis-specified (vision-language) actions during tasks’ execution are typical from model hallucinations. These can trigger deleterious consequences in terms of money, time, health, or even lives. Modularity in physical and virtual embodied reasoning systems is one way to imbue stability, robustness, and safety into embodied decision-making systems. AI agents that

composed sensing, perception, and action modules enable plug-and play optimization, reprogramming, and can improve embodiment and situatedness e.g. by combining a discrete high-level abstraction module (for reliably predicting and delineating environmental structure) with a continuous low-level decision-making module (that informs stable and safe world exploration). This hierarchical abstraction of planning in 3D worlds to sequential hybrids of discrete and continuous modules, informed by the power of existing foundation models together with disciplined combinatorial and convex optimization, and trained on large application-specific datasets, are crucial towards robustness and guaranteed safety in agent-centric multimodal decision-making environments.

II. RELATED WORK

In computer vision, perceptual engineering, and robotics, SLAM is the globally consistent solution concept that addresses the exploration problem in autonomous machines based on local observations. SLAM informs autonomous navigation, collision-free path-planning [10], volumetric configuration space mapping [14, 49], low-level control of mobile robots [75] More often than not, a full 3D reconstruction or *mapping* of the environment is a general step employed in situational understanding while simultaneously determining the position and orientation of the agent within the map in a so-called *localization* or tracking step. While 3D reconstruction [40] provides an all-encompassing representation schematic for further downstream tasks, for general autonomous embodied exploration tasks, an implicit representation [77] of the environment often suffices [66, 42, 59].

Over the two decades since a solution to the geometry-based SLAM problem was theoretically solved [23], SLAM has metamorphosed into broad applications including medical systems [81], augmented reality [50], visual surveillance, and beyond. These new developments tend to involve (i) creating an expressive capacity of environment models in a lower dimensional space or manifold — often called a *representation* problem; and (ii) analyzing the algorithmic performance of the inference systems that utilizes these representations.

A. Explicit and implicit representation schemes

In our viewpoint, new representation schemes can be broadly classified into explicit and implicit representation methods; and the algorithmic mechanisms for decision-making in these representation schemes vary in methods and their analysis — from the analysis of their computational performance to their robust and certifiable information-theoretic inference execution. Explicit methods directly map the perceptual volume of the environment to a discretized representation. These tend to store a large volume of measurement points, requiring at least the spatial volume of the scene in storing information. As such, they are very memory-prohibitive and are incapable of arbitrary 3D representation, such as tree branches [34]. Examples of these explicit representation methods include *occupancy grids* [27], *elevation maps* [47], *multi-level surface maps* [82], 3D grids of

equally-sized cubic ‘voxel’ volumes [71], *point clouds* [72], *signed distance functions* [67], or *triangular meshes* [26].

Point clouds store the full range of sensor information including free-space and occupied-space regions. For point clouds, the dynamic objects in the configuration space and the sensor noise innate to the environment are very difficult to directly model alongside the difficult objects classification it affords since it is a points-based representation. It is therefore computationally prohibitive from a memory storage standpoint in addition to the above reasons and its empty space representation.

Employing equally-sized cubic chunks of volume representation of 3D space, *voxels* [71, 61] discretize the environmental map into grids which are often lower-bounded by the environment’s bounding boxes, irrespective of the mapped cell volume distribution. This voxel representations is heavily used in the point cloud library for 3D segmentation, filtering, and points clustering, for example. However, being explicit representations, these schemes are not readily amenable to obstacle-free, unmapped areas representation, and uncertainty quantification in many 3D planning tasks.

Elevation [47] and *multi-surface maps* [82], while seeking to avoid these voluminous representation disadvantages, are not good enough for most unmapped areas since they have a predetermined size. *Octomap* [34] is a proposed remedy that aims to alleviate the reliability, open-source, and efficient memory storage requirements for 3D representations. It works by utilizing a probabilistic representation to model obstacle-free, occupied, and unmapped areas by exploiting a hierarchical mechanism in an Octree [58] data structure that minimizes memory consumption. However, it is CPU-based and in making a generalist model that encompasses diverse mediums including terrestrial, extraterrestrial, and underwater applications, it struggles in representation capacity. In addition, human operators struggle to interpret it, particularly when the agent being controlled is out of line of sight [63]. None of these approaches fulfill the nimbleness and generalization expectation to new environments in their (dynamic) representation capacity.

In implicit representation settings, truncated signed distance fields (TSDFs) [17, 62] is an implicit signed-distance representation scheme that is fast to compute in itself, process sensor noise better, and can create human-readable mesh renderings of the environment. However, the mesh rendering stage lends itself to a computationally intensive protocol that makes real-time operations difficult for a robot or an otherwise agent. Another example is *VoxBlox* [63], an incremental mapping strategy, based on voxels, that encodes TSDFs directly out of computed Euclidean Signed Distance fields (ESDFs) to surface boundaries in the scene based on a *truncation radius*. It has the limiting assumption on the availability of a stereo or RGB-D map input and assumes the agent’s pose is known *a priori*. This is not often the case for most field agents as the range information may need to be computed/estimated in real-time. In addition, it does not lend itself easily to a GPU implementation.

All of the above classical representation schemes are

almost all CPU based, computationally prohibitive, require costly storage requirements, or preprocessing before a representation of the environment is generated.

B. The case for a ‘stateful’ representation

In robotics, augmented/mixed reality for embodied devices, conformal ensembles of protein structures [15], and molecular chemistry, autonomous exploration [10] together with (perceptual) planning [11, 73] are time-consuming computational processes that involve reasoning over environmental geometry, developing the semantics thereby via world modeling (that is, mapping information from raw sensors to a low-dimensional space of representations), and agent localization, before low-level control (of an internal state) is executed. Other than in filtering and state-estimation protocols [41], the internal state is ephemeral and requires constant reasoning over rich scenery, and environmental geometry in order to determine *actionable semantics* which can be leveraged with e.g. sparse information from traditional sensors for effective (perhaps hierarchical) control.

An agent’s internal state¹, estimated or measured from sparse sensors, while good enough for low-level control, are often not efficient for global spatial reasoning and planning. Seamless situational awareness in large viewpoint and lighting changes, despite occlusions, blur, or texture degradation is a prerequisite for efficient exploration, planning, and control.

The need to understand *internal dynamics* from *perceptual interaction* with the environment in addition to *data association* with information gathered from high-resolution sensors such as RGBs has become ever more timely. For example, how do we create effective three-layer architectures [28] that solve the classical sense-plan-act [9] problem efficiently in an environment-agnostic and autonomous exploration manner? New thinking in robotics and spatial AI planning is already leading researchers to address this problem in bits and pieces. However, much of the existing work in literature address the problem in task-specific settings such as in indoor environments [36, 37, 70] or limited outdoor environments [3]; other works isolate autonomous exploration from the planning problem, or combine a manifold of hand-engineered features from raw sensors with planning and/or control schemes.

We focus on the entire autonomous exploration, planning, and control pipeline in a manner that is agnostic to indoor or outdoor environments, bring forth new insights to the exploration and (incremental) planning design pipeline, establish novel data-driven control principles, test our proposed hypotheses on a diversity of simulation and real-world experiments, and establish general new thinking paradigms that we hope can guide other tinkerers and researchers in this domain.

At issue is the real-time autonomous behavioral acquisition by virtual and/or real-world agents and systems in a perceptual engineering [11, 51, 73], planning [49], and

¹We generally mean a mathematical representation of state in geometric form as an aggregation of a position and orientation of a chosen point on its surface.

control context [28]. While geometric states [41] may be sufficient for planning in environments with high-resolution sensors [10, 5], when the agent must observe a spatiotemporally diverse world, develop a model before planning and control, these approach is not always sufficient owing to the time-consuming nature of acquiring world models, and processing such maps across various representation granularity because the semantics of the world may have changed before plans get executed. Thus, the internal state of the agent often loses synchronization with the reality owing to this computational disparity.

Leading from the axiom that “the world is its own best model”, how can we get agents to acquire autonomous behavior in a manner consistent with precise reasoning from compact representations of *complex perceptual information*, reactively develop its internal state from a (possibly crude) world model, and refine its internal state as plans are passed onto its low-level controller. Accompanying this modality must be means of *precisely representing* invariant information across *scene events*, “*dynamization*” of the ensuing representation, and imbuing such dynamics with (i) well-rounded sensing, (ii) precise planning (and world mapping), and (iii) control capabilities.

Fundamental to autonomous exploration and hence behavioral acquisition is the capability for an agent to explore its environment safely, and in a collision-free manner. Doing this requires imbuing autonomous agents with the classic sense-plan-act (SPA) three-layer control architecture [28]: the agent must interpret raw (possibly multiple) sensor signals from its environment into a world map or model. This map is handed off to a planner which takes an initial and goal configuration and generates a (possibly coarse) global plan. A low-level controller then isolates the steady-state convergence and setpoint tracking problem based on the plan given to it and produces a point-wise trajectory-tracking path in the configuration space in which the agent domiciles.

In this paper, we focus on a complete pipeline in the SPA architecture starting with learning-based perception engineering from raw observational information from the environment. Much of the research thrusts and literature that have explored the SPA scheme [10, 5, 18, 22, 76, 46], often restrict their sensing solution to a fusion of high-resolution sensors (e.g. high-fidelity light detection and ranging sensors or LiDaRs) with low-resolution ones (such as machine-vision camera sensors) [36, 37, 70, 53]. In this work, we assume no access to high-fidelity sensors; the agent is equipped with information arising from pixels only. As such, the agent must learn a *meaningfully metricized representation* of the environment from its (multiple) sensors via strategic and computationally tractable environmental exploration; flow the information acquired from sensors to planners; and then execute a plan

Perception engineering as we consider it may include spatial reasoning from high-dimensional vision sensors, rich kinesthetic sensing, or tactile and haptic sensing. We restrict the scope of our investigation to vision-based sensing for an initial analysis. Today, representations of rich observa-

tional information is often processed with powerful graphical processing unit (GPUs) devices owing to their use of embedded single instruction multiple thread (SIMT) computing schemes. This has enabled powerful advances in spatial AI perception [45], video control from compact scene representations [60], and robotic manipulation from pixels alone [53] — all based on the power of geometric deep learning — aided by high-resolution lithographical fabrication of highly miniaturized integrated circuits (ICs) densely packed into small-form factor electronic circuit boards.

While the long promise of perception engineering [51] may finally appear on the horizon, the well-known Moore’s direct scaling laws from ICs’ miniaturization presents technological challenges since power density decreases as area reduces owing to a lesser known Dennard’s law [21]. In this sentiment, *we focus on accelerating algorithmic development by considering sparse representation of rich perceptual information, and considering a holistic perceptual dynamical system by building a toolbox that leverages mathematical modeling, control systems, robotics, perceptual technology, and neuroscience in a principled learning fashion.*

The perceptual representational aspects of our contribution is motivated by studies in visual integration and change direction that suggest that human visual experience is characterized by (i) change blindness i.e. we do not detect large change details across temporally correlated scenes; and (ii) inattention blindness, i.e. we only perceive and remember objects that receive focused attention [73]; ,

In emphasis, an agent must leverage its embodied vision and/or proprioceptive sensing devices to efficiently condense real world scenes, (including perhaps auditory signals), and temporal *events* into a “small-scale internal model” of extrinsic world information. This small-scale model must contain the possible “legal” actions within the agent’s “imagination” or action space, available alternatives for safe action execution in the world, conclude the best among possible alternatives, anticipate future events, and adaptively react to real-time events based on previous experiences [16].

It has been hypothesized that evolutionary constraints imply that the features of the physical world that have not been relevant for our survival are discarded or not processed [11]. In this regard, there has been a recent trend in the deep reinforcement learning (DeepRL) community on representation learning of endogenous states solely from visual observations and planing from that learned representation.

C. The case for learned states in C-Free automata

Learning compact representations from high-dimensional temporally-varying signal and observation spaces is a key challenge in computer vision [29], signal processing [85], and control-from-pixels [30, 29, 31]. Learned representations from neural architectures such as convolution [52], recurrent [33], and multilayer perceptron networks, provide *ad-hoc* prediction efficacy in classification and statistical learning tasks. However, they tend to be overparameterized for reliable control in large observation spaces. As a result,

this slakes their performance in long-range prediction and control tasks.

In this light, data-driven control-from-pixels has remained elusive for learning agents in multiple domains including simulated reality [32], partially-observed environments [78, 79], and in challenging robotics tasks [48] with assured transients. This is in part due to the lack of separation between model and control in direct data-driven (model-free) decision-making frameworks: the best data fit that is identified may not inform the final decision-making task. In general, methods for learning decisions in these model-free schemes

- lack compression because they employ deep learning modules, which are well known to be data-inefficient;
- tend to be noisy given the squashed representation structure; and
- lack homomorphic injective properties to the observation world as with Port Hamiltonian systems e.g. [84].

The conventional wisdom in representation learning settings is to aggregate learned *mid-level* representations from a complex observation world [29] in an autoencoder (sometimes coupled with decoder) mechanism; this is then employed to synthesize a policy within the *arbitrarily-sized* learned latent representation. We thus concentrate our state representation efforts around model-based decision-making systems (from neural networks) that

- filter the data and minimizes variance;
- allow incorporating side information such as equivariance priors, inductive biases;
- allow efficiently incorporating exotic online learning properties including dual identification and control into the modeling process [86]; and
- create representation structures amenable to existing powerful design methods such as semidefinite programming, nonlinear observers, Lie group time integrations [38], and sliding mode control [83].

In our quest towards data-driven decision-making for complex dynamical systems, using a model for learning is helpful in avoiding the foregoing issues. Models allow capturing parametric, interpretable, invariant, long-range dynamics [25] for instance, characterized by a complete, dense model set [13].

D. Interpretable spatial representation via convex decomposition

- 1) Removing exact convex decomposition constraints:
- 2) Convex decomposition via volume-based convex decomposition:

III. METHODS

A. Data Preprocessing

- What datasets did we use?
- Why did we pick those datasets?
- Which sensor info did we include in our dataset – IMU, LiDAR, Depth, RGB etc.?
- What preprocessing did we do?
- How did we store the dataset on memory and why?

IV. SELF-SUPERVISED CONVEX SUPERVISION: TRAINING LOSSES

Our

V. NUMERICAL RESULTS

VI. CONCLUSION

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